

Lecture 9: Derived Distributions

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- 1 Finding the distribution of $Y = g(X)$::: `{.callout-warning icon=false}## FactCDF method: compute  $F_Y(y) = P(g(X) \leq y)$  directly. .... . ::: {.callout-warning icon=false}## Theorem: Density transformation formulaFor monotonic, differentiable  $g$ :`

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

- :::—## **Worked example: $Y = X^2$ for $X \sim N(0,1)$** 1
1.1 Reading & HomeworkReading: §4.1 (*Bertsekas & Tsitsiklis, Introduction to Probability*).Due
today: Homework 4 (Gradescope). 2

—title: “Lecture 9: Derived Distributions”author: “Jamie Haddock”format: revealjs: output-file: Lecture9_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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- :::—## **Worked example: $Y = X^2$ for $X \sim N(0,1)$**

```
from scipy.stats import chi2X = rng.normal(size=200000)Y = X**2ys = np.linspace(0.001, 8, 400)plt.hist(Y, bins=ys)
```

—# Case Study: Randomized Matrix MultiplicationWrite $AB = \sum_{i=1}^m a_i b_i^\top$ (a sum of rank-one terms, columns a_i of A / rows $b_i^\top$ of B). Sample a single index I and rescale::: `{.callout-note icon=false}## Definition: Sampling estimator for  $AB$`

$$Z = \frac{1}{p_I} a_I b_I^\top, \quad p_i \propto \|a_i\| \|b_i\|.$$

Unlike a named family, p is **derived** directly from the data. We’ll find the *best* choice of p in Week 9 (Lecture 16):::

```
n, m, p_dim = 6, 40, 5A = rng.normal(size=(n, m))B = rng.normal(size=(m, p_dim))true_AB = A @ Bnorms =
```

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